

CHAPTER 2

Introducing economic primitives

The strength of the Anthropic Economic Index lies in showing not only how much AI is used, but how it is used. In prior reports, we showed which tasks Claude is used for, and how people collaborate with Claude. These data have enabled external researchers to analyze labor market shifts (e.g., [Brynjolfsson, Chandar & Chen, 2025](#)).

In this edition of the Anthropic Economic Index, we expand the breadth of data available to external researchers by providing insights on five economic “primitives”, by which we mean simple, foundational measures of the ways that Claude is used, which we generate by asking Claude to answer specific questions about the anonymized transcripts in our sample. Some of our primitives encompass several such questions, and others use a single indicator.

Because AI capabilities are advancing so rapidly and the economic effects will be unevenly experienced, we need a breadth of signals to uncover not just how Claude is used but also to inform what impact this technology will have.

Dimensions of AI use that matter for economic impacts

This report introduces five new economic primitives beyond the one we already measure, collaboration patterns (whether users automate or augment their tasks with Claude). These primitives capture five dimensions of a human-AI conversation: 1) task complexity, 2) human and AI skills, 3) work, coursework or personal use case, 4) the AI’s level of autonomy, and 5) task success (see Table 2.1). AI autonomy captures something different from our existing automation/augmentation distinction. For example, “Translate this paragraph into French” is high automation (directive, minimal back-and-forth) but low AI autonomy (the task requires little decision-making from Claude).

Economic primitives overview

Task complexity	Human time estimate	Estimate how many hours a competent professional would need to complete the tasks done by the Assistant. Assume they have: • The necessary domain knowledge and skills • All relevant context and background information • Access to required tools and resources • No access to AI tools to assist with the work
	Human with AI time estimate	Estimate how many minutes the User spent completing the tasks in the prompt with the Assistant. Consider: • Number and complexity of User messages • Time reading Assistant's responses • Time thinking and formulating questions • Time reviewing outputs and iterating • Realistic typing/reading speeds • Time implementing suggestions or running code outside of the conversation (only if directly relevant to the tasks)
	Multitasking	Did the User multitask in this conversation? Choose from these options: • Yes: the User was working on multiple tasks over the course of the conversation • No: the User was working on a single task over the course of the conversation
Human and AI skills	Human ability to complete task alone	Could the User have completed this task by themselves? Choose from these options: • Yes: the User would have been able to complete the task without the Assistant, even if it would have taken more time • No: the User would not have been able to complete the task without the Assistant, even with more time
	Human education years	Estimate how many years of formal education someone would need to understand the User prompts in this conversation. Your answer should be a single number out of the discrete numbers ranging from 0-20.
	AI education years	Estimate how many years of formal education someone would need to understand the Assistant responses in this conversation. Your answer should be a single number out of the discrete numbers ranging from 0-20.
Use case	Work vs. coursework vs. personal	Analyze whether the conversation between the User and the Assistant primarily focuses on work, coursework or personal use. Analyze the use case according to these categories: • Work: professional use to accomplish tasks that are part of the User's job • Coursework: use to help the User complete coursework in educational contexts • Personal: use for any domain that is not work or coursework
AI autonomy		Estimate how much autonomy the Assistant had to make decisions in this conversation (a discrete number ranging from 1-5, where 1 is none and 5 is extreme).
Task success		Did the Assistant complete the task provided by the User successfully? Choose from these options: • Yes: the Assistant completed the task provided by the User successfully • No: the Assistant did not complete the task provided by the User successfully

Table 2.1: Economic primitives added in this report.

The table shows the new economic primitives added in this report, beyond collaboration patterns (automation/augmentation) from prior reports. The first column shows the primitive category, the second column the name of the primitive, and the third column the operationalization of the primitives as the prompts provided to Claude which we use a classifier to map conversations to primitives. See online appendix at <https://huggingface.co/datasets/Anthropic/EconomicIndex> for full prompt texts.

Task complexity captures that tasks can vary in their complexity, including how long they take to complete and how difficult they are. A “debugging” task in O*NET could refer to Claude fixing a small error in a function or comprehensively refactoring a codebase—with very different implications for labor demand. We measure complexity through estimated human time to complete tasks without AI, time spent completing tasks with AI, and whether users handle multiple tasks within a single conversation.

Human and AI skills address how automation interacts with skill levels. If AI disproportionately substitutes for tasks requiring less expertise while complementing higher-skilled work, it could be another form of skill-biased technical change—increasing demand for highly skilled workers while displacing lower skilled workers. We measure whether users could have completed tasks without Claude, and the years of education needed to understand both user prompts and Claude’s responses.

Use case distinguishes professional, educational, and personal use. Labor market effects most directly follow from workplace use, while educational use may signal where the future workforce is building AI-complementary skills.

AI autonomy measures the degree to which users delegate decision-making to Claude. Our latest report documented rising “directive” use where users delegate tasks entirely. Tracking autonomy levels—from active collaboration to full delegation—helps forecast the pace of automation.

Task success measures Claude’s assessment of whether Claude completes tasks successfully. Task success helps assess whether tasks can be automated effectively (can a task be automated at all?) and efficiently (how many attempts would it take to automate a task?). That is, task success matters for both the feasibility and the cost of automation labor tasks.

Selecting and validating the new measures

The new dimensions of AI use captured in our data were informed by our recent work on the [productivity effects of Claude](#), feedback we received from external researchers, recent literature on AI’s economic impact through the lens of human capital and expertise ([Vendraminell et al., 2025](#)), and